文献汇报

高铭阳

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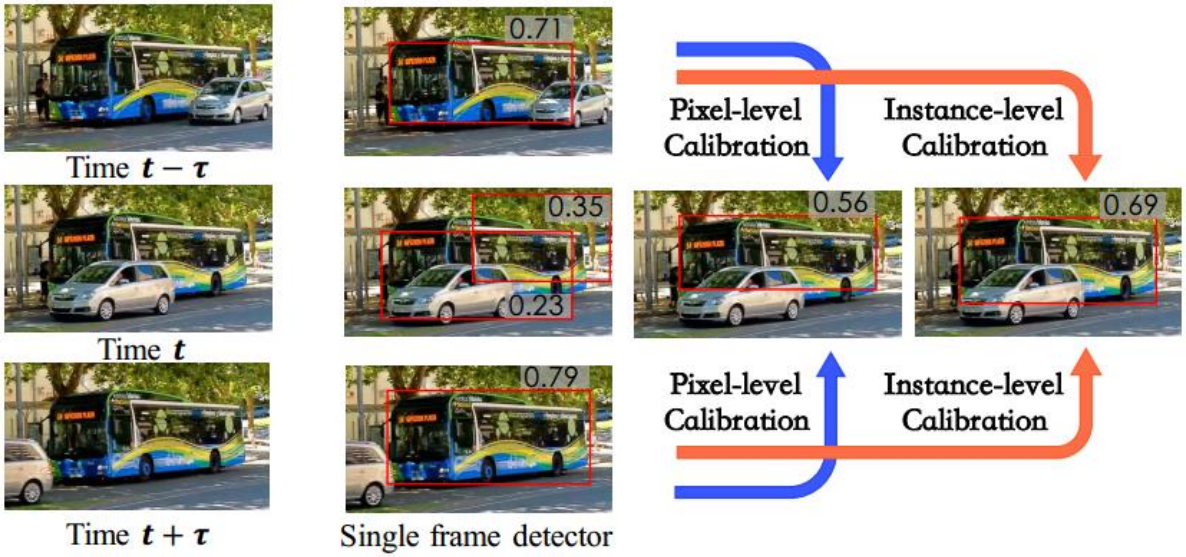
Fully Motion-Aware Network for Video Object Detection

Shiyao Wang[1], Yucong Zhou[2], Junjie Yan[2], and Zhidong Deng[1]

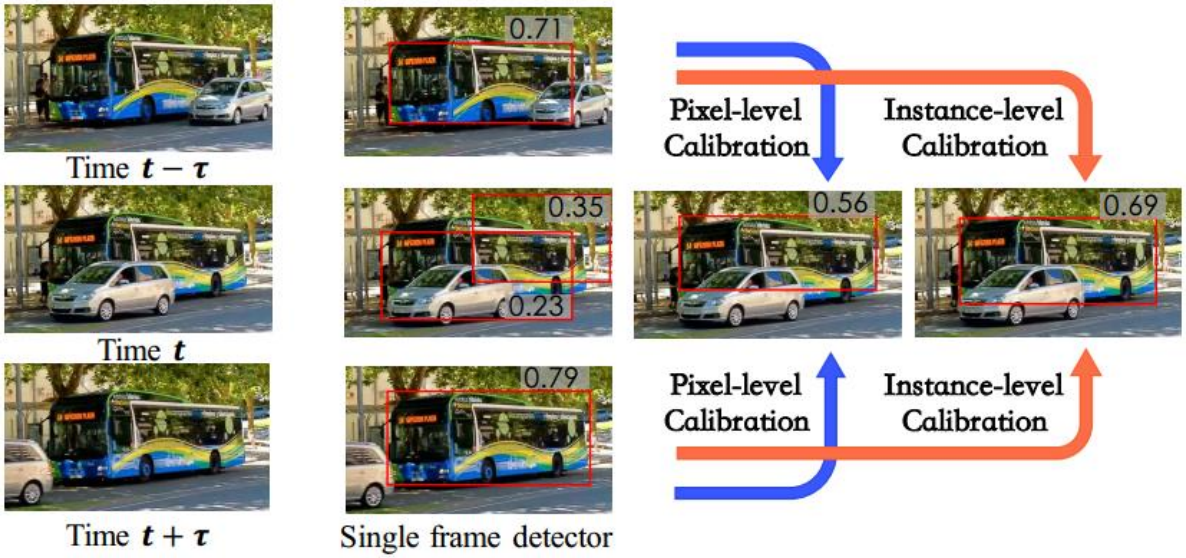
[1] State Key Laboratory of Intelligent Technology and Systems
Beijing National Research Center for Information Science and Technology
Department of Computer Science, Tsinghua University, Beijing 100084, China
[2] SenseTime Research Institute
sy-wang14@mails.tsinghua.edu.cn, zhouyucong@sensetime.com
yanjunjie@sensetime.com, michael@tsinghua.edu.cn



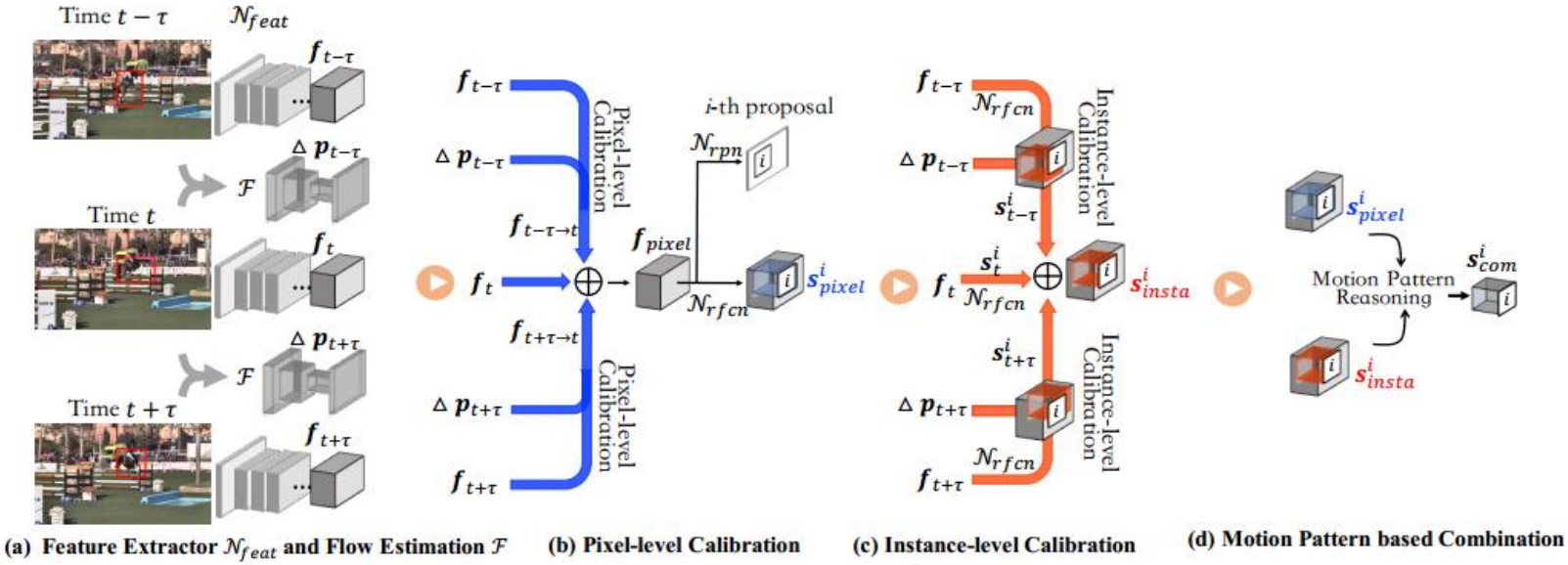
介绍

视频的帧经常由于运动模糊或散焦等原因退化，直接想到从时间空间的相关性和临近帧中来获取信息。FAGA尝试时间相关性来从临近帧聚合特征，这是像素级的校准。问题是目标变动剧烈（尤其是遮挡）就不精确。

本文除了像素级校准，还提出了实例级校准，能估计每一个目标随时间的运动方式来精确地聚合特征。相比于像素级校准，实例级校准对较大的时间外观变化更鲁棒。而且，两种校准方式能根据不同的运动模式协同工作。像素级适合非刚性变换，尤其是小动物的运动。本文采用了一个运动模式解释模型，如果运动模式更像是非刚性运动而且无遮挡，那最终结果更基于像素级校准。反之，基于实例级校准。以上模型都集成在一个联合框架上可端到端训练。



方法



(a) Feature Extractor $\mathcal{N}_{feat}$ and Flow Estimation $\mathcal{F}$ (b) Pixel-level Calibration (c) Instance-level Calibration (d) Motion Pattern based Combination

可以从上图看出MANET的流程

1、通过N-feat提取特征，用Flow的方法生成非关键帧的特征。

2、像素级校准

3、实例级校准

4、结合

Pixel-level Calibration

$$\begin{aligned} \boldsymbol{f}_{t-\tau} &= \mathcal{N}_{feat}(\boldsymbol{I}_{t-\tau}) \\ \boldsymbol{f}_{t-\tau\to t} &= \mathcal{W}(\boldsymbol{f}_{t-\tau}, \mathcal{F}(\boldsymbol{I}_{t-\tau}, \boldsymbol{I}_t)) \end{aligned} \tag{1}$$

where $\boldsymbol{f}_{t-\tau}$ denotes feature maps extracted by $\mathcal{N}_{feat}$ and $\boldsymbol{f}_{t-\tau\to t}$ is the warped features from time $t-\tau$ to time t. The warping operation $\mathcal{W}$ is implemented by bi-linear function which is applied on each location for all the feature maps. It projects a location $\boldsymbol{p}+\Delta\boldsymbol{p}$ in the nearby frame $t-\tau$ to the location $\boldsymbol{p}$ in the current frame. We formulate it as:

$$\begin{aligned} \Delta\boldsymbol{p} &= \mathcal{F}(\boldsymbol{I}_{t-\tau}, \boldsymbol{I}_t)(\boldsymbol{p}) \\ \boldsymbol{f}_{t-\tau\to t}(\boldsymbol{p}) &= \sum_{\boldsymbol{q}} G(\boldsymbol{q}, \boldsymbol{p}+\Delta\boldsymbol{p})\boldsymbol{f}_{t-\tau}(\boldsymbol{q}) \end{aligned} \tag{2}$$

where $\Delta\boldsymbol{p}$ is the output of flow estimation at location $\boldsymbol{p}$. $\boldsymbol{q}$ enumerates all spatial locations in the feature maps $\boldsymbol{f}_{t-\tau}$, and $G(\cdot)$ denotes bi-linear interpolation kernel as follow:

$$G(\boldsymbol{q}, \boldsymbol{p}+\Delta\boldsymbol{p}) = max(0, 1-||\boldsymbol{q}-(\boldsymbol{p}+\Delta\boldsymbol{p})||) \tag{3}$$

After obtaining calibrated features of nearby frames, we average these features as the low-level aggregation for the updated reference features:

$$\boldsymbol{f}_{pixel} = \frac{\sum_{j=t-\tau}^{t+\tau} \boldsymbol{f}_{j\to t}}{2\tau+1} \tag{4}$$

Instance-level Calibration

First, we utilize the RoI pooling operation to generate the pooled features $\boldsymbol{m}^i_{t-\tau}$ of the i-th proposal at location $(x^i_t, y^i_t, h^i_t, w^i_t)$:

$$\boldsymbol{m}^i_{t-\tau} = \phi(\mathcal{F}(\boldsymbol{I}_{t-\tau}, \boldsymbol{I}_t), (x^i_t, y^i_t, h^i_t, w^i_t)) \tag{5}$$

where $\phi(\cdot)$ indicates the RoI pooling [6] and $\mathcal{F}(\boldsymbol{I}_{t-\tau}, \boldsymbol{I}_t)$ is the flow estimation produced by shared FlowNet in Section 3.2. RoI pooling uses max pooling to convert the features inside any valid region of interest into a small feature map with fixed spatial extent.

Then regression network $R(\cdot)$ is exploited to estimate the movement of the i-th proposal between the frame $t-\tau$ and t according to the $\boldsymbol{m}^i_{t-\tau}$:

$$(\Delta^i_{x_{t-\tau}}, \Delta^i_{y_{t-\tau}}, \Delta^i_{w_{t-\tau}}, \Delta^i_{h_{t-\tau}}) = R(\boldsymbol{m}^i_{t-\tau}) \tag{6}$$

where $(\Delta^i_{x_{t-\tau}}, \Delta^i_{y_{t-\tau}}, \Delta^i_{w_{t-\tau}}, \Delta^i_{h_{t-\tau}})$ is relative movements and $R(\cdot)$ is implemented by a fully connected layer. The remaining problem is how to design proper supervisions for learning the relative movements. Since we have the track-id of each object within a video, we are able to generate the relative movements in terms of the ground-truth bounding boxes. We believe that the proposals should have consistent movement with the ground-truth objects. Thus, the above regression target is assigned the ground-truth box movement if the proposal overlaps with a ground-truth at least by 0.5 in intersection-over-union (IoU). In other

Instance-level Calibration

Once we obtain the relative movements, we are able to calibrate the features across time and aggregate them to enhance the feature of the current frame. The proposal of frame $\boldsymbol{I}_{t-\tau}$ can be inferred as:

$$\begin{aligned} x^i_{t-\tau} &= \Delta^i_{x_{t-\tau}} \times w^i_t + x^i_t \qquad & y^i_{t-\tau} &= \Delta^i_{y_{t-\tau}} \times h^i_t + y^i_t \\ w^i_{t-\tau} &= exp(\Delta^i_{w_{t-\tau}}) \times w^i_t \qquad & h^i_{t-\tau} &= exp(\Delta^i_{h_{t-\tau}}) \times h^i_t \end{aligned} \tag{7}$$

Based on the estimated proposal locations for nearby frames, the aggregated feature of the i-th proposal can be calculated as:

$$\boldsymbol{s}^i_{insta} = \frac{\sum_{j=t-\tau}^{t+\tau} \psi(\boldsymbol{s}_j, (x^i_j, y^i_j, h^i_j, w^i_j))}{2\tau + 1} \tag{8}$$

where $\boldsymbol{s}_j$ denotes the neighboring score maps, ψ indicates position-sensitive pooling layer introduced by [2], and $\boldsymbol{s}^i_{insta}$ is the instance-level calibrated feature of the i-th proposal.

Motion Patten Reasoning and Overall Learning Objective

最后，运用运动模式推理来决定怎样结合这些不同的经过校准的features。问题的关键在于怎样度量运动模式的非刚性。

直观的说，bounding box的长宽比在运动中随着时间是急剧变化的，因此它的运动模式更像是非刚性。使用的是质心差异来表达bounding box长宽比的变化率。

$$\delta(\frac{x_t^i}{y_t^i}) = (\frac{x_{t+1}^i}{y_{t+1}^i} - \frac{x_{t-1}^i}{y_{t-1}^i})/2$$
$$p_{nonri}^i = \frac{\sum_{j=t-\tau+1}^{t+\tau-1} \delta(\frac{x_j^i}{y_j^i})}{2\tau - 1} \tag{9}$$

where p_{nonri}^i is the motion pattern descriptor for the i-th proposal. The corresponding proposals in the nearby frames can be obtained from Section 3.3.

Additionally, occlusion is another important factor when combining these two calibrations. We exploit the visual feature within the proposal to predict the probability of the object being occluded:

$$p_{occlu}^i = R(\phi(\boldsymbol{f}_t, (x_t^i, y_t^i, h_t^i, w_t^i))) \tag{10}$$

where $R(\cdot)$ is also implemented by a fully connected layer and p_{occlu}^i is the probability of occlusion for the i-th proposal. Notice that Equation 10 is similar to Equation 6, but Equation 6 uses motion features from FlowNet to regress movements while Equation 10 adopts visual features to predict occlusion. It is mainly due to the fact that occlusion is more related to appearance.

Considering these two factors, we use learnable soft weights to combine the two calibrated features:

$$\boldsymbol{s}_{com}^i = \boldsymbol{s}_{insta}^i \times \alpha(\frac{p_{occlu}^i}{p_{nonri}^i}) + \boldsymbol{s}_{pixel}^i \times (1 - \alpha(\frac{p_{occlu}^i}{p_{nonri}^i})) \tag{11}$$

实验

Feature Extractor	ResNet-101				
Methods	(a)	(b)	(c)	(d)	(e)
multi-frame feature aggregation?		√	√	√	√
Pixel-level Calibration?			√		√
Instance-level Calibration?				√	√
mAP(%)	73.6	73.4 $\downarrow_{0.2}$	76.5 $\uparrow_{2.9}$	77.1 $\uparrow_{3.5}$	**78.1** $\uparrow_{\mathbf{4.5}}$
mAP(%)(slow)	81.8	83.8 $\uparrow_{2.0}$	85.0 $\uparrow_{3.2}$	85.5 $\uparrow_{3.7}$	**86.9** $\uparrow_{\mathbf{5.1}}$
mAP(%)(medium)	71.3	75.7 $\uparrow_{4.4}$	74.9 $\uparrow_{3.6}$	76.1 $\uparrow_{4.8}$	**76.8** $\uparrow_{\mathbf{5.5}}$
mAP(%)(fast)	52.2	45.2 $\downarrow_{7.0}$	56.6 $\uparrow_{4.4}$	55.4 $\uparrow_{3.2}$	**56.7** $\uparrow_{\mathbf{4.5}}$

实验

Methods	airplane	antelope	bear	bicycle	bird	bus	car	cattle	dog	d_cat	elephant	fox	g-panda	hamster	horse	lion
R-FCN[2]	90.5	80.1	83.0	69.6	73.4	72.4	57.2	62.5	69.0	81.6	77.3	85.0	80.7	87.0	72.5	41.6
TPN+LSTM[13]	84.6	78.1	72.0	67.2	68.0	80.1	54.7	61.2	61.6	78.9	71.6	83.2	78.1	91.5	66.8	21.6
D (& T loss)[5]	89.4	80.4	83.8	70.0	71.8	82.6	56.8	71.0	71.8	76.6	79.3	89.9	83.3	91.9	76.8	57.3
DFF[31]	84.6	82.1	84.1	67.1	71.1	76.1	56.5	67.8	65.0	82.3	76.3	87.8	81.9	91.3	70.3	47.7
FGFA[30]	89.4	85.1	83.9	69.8	73.5	79.0	60.6	70.7	72.5	84.3	79.9	89.8	81.0	93.3	72.3	50.5
MANet	90.1	87.3	83.4	70.9	73.0	75.6	62.0	74.0	73.3	85.3	79.6	91.6	83.5	96.5	74.5	70.5
TCN [15]	72.7	75.5	42.2	39.5	25.0	64.1	36.3	51.1	24.4	48.6	65.6	73.9	61.7	82.4	30.8	34.4
TCNN[14]	83.7	85.7	84.4	74.5	73.8	75.7	57.1	58.7	72.3	69.2	80.2	83.4	80.5	93.1	84.2	67.8
D (& T loss)(τ= 1)[5]	90.2	82.3	87.9	70.1	73.2	87.7	57.0	80.6	77.3	82.6	83.0	97.8	85.8	96.6	82.1	66.7
MANet (+[8])	88.7	88.4	86.9	71.4	73.0	78.9	59.3	78.5	77.8	90.6	79.1	96.3	84.8	98.5	77.4	75.5

Methods	lizard	monkey	motor	rabbit	r-panda	sheep	snake	squirrel	tiger	train	turtle	watercraft	whale	zebra	mAP(%)
R-FCN[2]	78.0	52.2	81.2	66.6	81.5	57.3	70.5	53.1	90.8	82.3	79.1	64.6	75.0	91.2	73.6
TPN+LSTM[13]	74.4	36.6	76.3	51.4	70.6	64.2	61.2	42.3	84.8	78.1	77.2	61.5	66.9	88.5	68.4
D (& T loss)[5]	79.0	54.1	80.3	65.3	85.3	56.9	74.1	59.9	91.3	84.9	81.9	68.3	68.9	90.9	75.8
DFF[31]	76.5	45.7	78.1	62.8	77.8	55.8	74.5	50.5	90.2	81.7	77.9	65.8	66.2	89.5	72.8
FGFA[30]	80.8	52.3	83.0	72.7	84.0	57.8	77.1	55.8	91.9	83.8	83.3	68.7	75.9	91.1	76.5
MANet	82.0	54.4	81.6	67.0	89.3	73.3	77.4	54.3	91.9	82.9	80.3	69.3	75.4	92.4	**78.1**
TCN[15]	54.2	1.6	61.0	36.6	19.7	55.0	38.9	2.6	42.8	54.6	66.1	69.2	26.5	68.6	47.5
TCNN[14]	80.3	54.8	80.6	63.7	85.7	60.5	72.9	52.7	89.7	81.3	73.7	69.5	33.5	90.2	73.8
D (& T loss)(τ= 1)[5]	83.4	57.6	86.7	74.2	91.6	59.7	76.4	68.4	92.6	86.1	84.3	69.7	66.3	95.2	79.8
MANet(+[8])	84.8	55.1	85.8	76.7	95.3	76.2	75.7	59.0	91.5	81.7	84.2	69.1	72.9	94.6	**80.3**